ANSHUL PAL

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TECHNICAL SKILLS

Languages:	Go, C, Python, JavaScript, TypeScript, SQL (PostgreSQL), HTML5, CSS3
Core CS & Systems:	Distributed Systems (Raft Consensus), Operating Systems, POSIX Syscalls, Concurrency & Locks, Linux Internals, Data Structures & Algorithms
Backend & Frameworks:	Node.js, Express, Next.js (App Router), React, RESTful APIs, TCP Sockets, Prisma ORM, Web Audio API
Developer Tools & Cloud:	Git, GitHub, Linux (Bash, WSL2), Cgroups v2, Namespaces, Make, GCC, Render, Supabase

TECHNICAL PROJECTS

Miniraft – Distributed KV Store & Lock Manager github.com/anshul1729-in/miniraft
Go, TCP RPC, Distributed Consensus, Concurrency *Oct 2026*

- Architected a fault-tolerant distributed Key-Value store and Lock Manager in Go implementing the complete Raft Consensus Algorithm from scratch over raw TCP RPCs.
- Engineered leader election with randomized election timers (300–600ms) and strict quorum validation ($\lfloor N/2 \rfloor + 1$), completely preventing split-brain states across cluster network partitions.
- Implemented two-phase commit log replication (`AppendEntries`), achieving guaranteed state machine consistency and surviving abrupt `kill -9` leader failure in ~400ms with zero committed log loss.
- Formulated idempotent mutation handling via client-generated `req_id` deduplication state tables, preventing duplicate state machine execution during network retries and leader failovers.

Minishell – Custom POSIX Command Shell github.com/anshul1729-in/minishell
C, Linux Syscalls, Process Management, POSIX API *Oct 2026*

- Engineered a custom POSIX-compliant shell in C orchestrating process lifecycles via low-level kernel syscalls (`fork()`, `execve()`, `waitpid()` with `WUNTRACED` and `WNOHANG`).
- Implemented an arbitrary-depth multi-stage pipeline executor (`cmd1 | cmd2 | cmd3`) by dynamically rewiring per-process File Descriptor tables using `pipe()` and `dup2()`.
- Built comprehensive I/O redirection (`<`, `>`, `2>&1`) and terminal session job control (`setpgid()`, `tcsetpgrp()`) to isolate foreground and background process groups.
- Designed asynchronous signal multiplexing for `SIGINT` (Ctrl+C), `SIGTSTP` (Ctrl+Z), and `SIGCONT`, maintaining an interactive process table for job control (`jobs`, `fg`, `bg`).

NeonWave – Algorithmic Music Recommendation & Streaming Platform github.com/anshul1729-in/Music-app
Node.js, Express, PostgreSQL, React, Web Audio API *Jun 2026*

- Designed and deployed an algorithmic recommendation backend implementing exponential time-decay scoring ($\lambda = \exp(-\Delta t / 7d)$) over granular telemetry (play complete, skip, progress).
- Constructed session transition Markov models and playlist co-occurrence collaborative filtering using complex PostgreSQL Common Table Expressions (CTEs) and window functions (`LEAD`).
- Engineered an in-memory TTL caching layer (60,000ms) for recommendation feeds, eliminating redundant analytical DB queries and preventing thundering herd bottlenecks.
- Automated production cloud deployment with infrastructure-as-code configuration (`render.yaml`) connecting managed PostgreSQL and service environments.

Minicontainer – Linux Container Runtime from Scratch github.com/anshul1729-in/minicontainer
Go, Linux Namespaces, Cgroups v2, Syscalls *Oct 2026*

- Developed a lightweight container runtime in Go directly interacting with Linux kernel primitives via Go `syscall`, achieving isolated process execution without Docker or containerd.
- Isolated process trees, mounts, and network interfaces using Linux `CLONE_NEWPID`, `CLONE_NEWNS` (`MS_PRIVATE`), `CLONE_NEWUTS`, and `CLONE_NEWNET` namespaces.
- Enforced strict hardware quotas via Linux **Cgroups v2** (`/sys/fs/cgroup`), capping container memory at 50MB (swap disabled) and throttling CPU utilization to a 50% core quota.

WORK EXPERIENCE

PropertyLay Jun 2025 – Sep 2025
Freelance Software Engineer *New Delhi, India*

- Architected and delivered a responsive commercial real estate web platform using Next.js (App Router), React, and Tailwind CSS for dynamic property portfolios.
- Engineered client-side routing, modular UI component hierarchies, and performant dynamic lead capture forms integrated with input validation pipelines.
- Optimized asset delivery, metadata, and Core Web Vitals, achieving sub-second Largest Contentful Paint (LCP) and zero-layout-shift UI transitions.

EDUCATION

Amity University Expected 2027
Bachelor of Computer Applications (BCA) – Computer Science *Noida, India*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming.